

Generation Interconnection System Impact Study Report

For Queue Project Q20-1494

Permanente 138 kV

June 16, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1494. The Interconnection Customer proposes a 53.1 MW solar photovoltaic facility interconnecting at the Permanente 138 kV substation in the Pennsylvania area. The request was received on June 11, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 546 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,035,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of CARBON CTR 500-345 kV Auto, the SHAWVILLE STATION 500-345 kV XFMR loads to 109.5% of its 1,472 MVA rating. Project DFAX: 8.7%.

For the loss of NEW MELONES - ARMSTRONG 345 kV, the SIBLEY - INDIANA UNIVERSITY OF PENNSYLVANIA 138 kV loads to 106.4% of its 182 MVA rating. Project DFAX: 5.2%.

For the loss of FORT HUMBUGH - ORCHARD RD 500 kV L, the FORT HUMBUGH 500/345 kV TX loads to 106.0% of its 1,756 MVA rating. Project DFAX: 5.1%.

FT HUMBUGH SUBSTATION - LITTLE CHUTE 345 kV: 103.2% loading (2,008 MVA) following the outage of the CORONET - FORT HUMBUGH SUBSTATION 500 kV; DFAX = 5.8%.

The SCOTT - SHAWVILLE STA 345 kV is overloaded to 102.0% of its 1,938 MVA emergency rating for the loss of the ORCHARD RD - SHAWVILLE STA 500 kV. The project contributes 8.8% DFAX (4.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SHAWVILLE STATION 500-345 kV XFMR	Replace transformer with higher-rated unit	\$46,546,000	2% (\$1,114,000)
SIBLEY - INDIANA UNIVERSITY OF PENNSYLVANIA 138 kV	Reconductor line	\$7,286,000	100% (\$7,286,000)
FORT HUMBUGH 500/345 kV TX	Replace transformer with higher-rated unit	\$30,985,000	2% (\$605,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LITTLE CHUTE - FORT HUMBUG 345 kV line	Reconductor line	\$34,535,000	1% (\$458,000)
SCOTT - SHAWVILLE STA 345 kV	Reconductor line	\$33,727,000	2% (\$572,000)
	Total allocated to project		\$10,035,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.